SCHOOL OF COMPUTINNG, ENGINEERING AND BUILT ENVIRONMENT

Dept. of Computing

SESSION: 2025-2026 **DIET: 1st**

Level: M

Software Engineering and DevOps (MMI330704)

Module Leader: Dr Sanaullah Manzoor

COURSEWORK 2 (Group-based)

Requirements and Case Study/Forms

Total Marks: 100

Minimum Pass Marks: 50

Deadline: Monday 20th April 2026 15:00

(50% of overall mark)

Coursework 2
DevOps from Code to Application Deployment

In this coursework you will implement the complete DevOps pipeline using all of the tools you have learned in this module with a focus on continuous deployment. The goal is to develop a system similar to the diagram below:

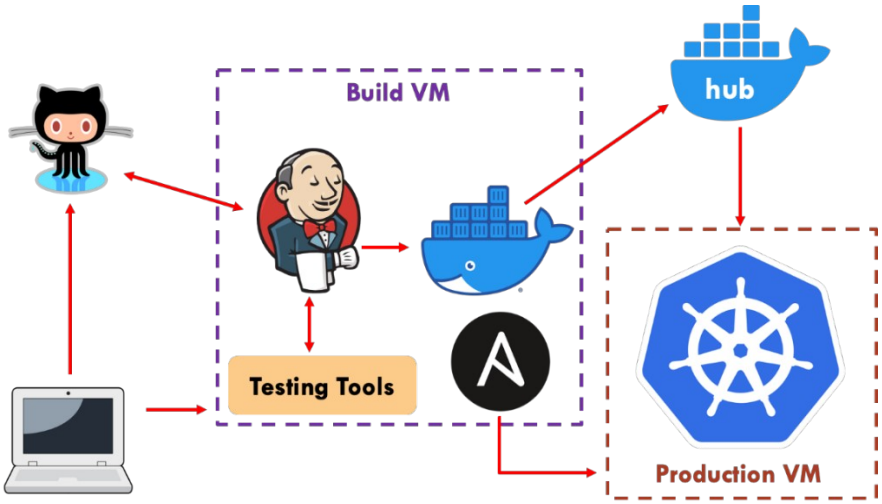


Figure 1 - DevOps Pipeline Overview

The coursework consists of these 3 main components:

- **Written Component (50%)** (MS Word Document) - A report focused on software tools and DevOps transformation
- **Practical Component (40%)** (MS Word Document with Codes and screenshots)- Building an application, Preparing the Production Server, Configuring CD Pipeline
- **Individual Report (10%) MS Word Document** - Explain and mark own and team members' contributions, success & weakness of the project

Submission:

There will be two submission portals on GCULearn:

- Group Report - Submitted via Turnitin Assignment
- Practical Evidence - Submitted via Turnitin Assignment.
- Individual Report- Submitted via Turnitin Assignment.
  The video link must be included in your Supplementary_Evidence_FULLNAME.doc. The doc must be submitted to coursework 2 practical component submission portal.

Both components of this coursework should be submitted to GCULearn no later than:

Monday 20th April 2026 at 13:00

Please note that **only one submission can be made** so please make sure you are happy before you submit - **Submissions are final!**

This is a group coursework, so ALL members of a group must participate. Student engagement will be monitored. Poor engagement and/or contributions will result in a poor mark. You are expected to work on your coursework outside timetabled lab times as well as during lab times.

Plagiarism and Collusion

Note: 'Here at Glasgow Caledonian University, we want you to understand that practising academic integrity means keeping our values in mind when you are doing your academic work, particularly when using the work of others.'

Please read further about using GenAI in your studies:
https://www.gcu.ac.uk/currentstudents/essentials/onlinelearning/assessmentandintegrity

Please note that **only one submission can be made** so please make sure you are happy before you submit - **Submissions are final!**

1 Written Component (MS Word Document)- 50 Marks

This report will focus on two main themes which are outlined below.

This should be a formal report and should be **a maximum of 1000 words** (+/- 10%) excluding references, captions, contents and cover page. Exceeding the word limit will result in a mark deduction of 5% points.
Below you will find details of the various components of this report along with a marking Rubric that can be found in **Error: Reference source not found**.

1.1 Part 1 - Software Tools (300 words)-*25 Marks*

DevOps attempts to address the challenges common in traditional software development environments through a number of cultural and technical practices. Discuss the following software tools providing detail on their functionality and how they contribute to a DevOps environment.

a) Ansible in the context of Configuration Management (CM) (Ansible is used in a number of contexts so make sure you focus on CM)
b) Kubernetes

1.2 Part 2 - DevOps Transformation (700 words)- *25 Marks*

You have just joined a company who provide a data analysis application to the financial sector. In recent years it has become increasingly difficult to keep up with the demands of the customers in terms of their new feature requests. In attempting to roll out new features to meet this demand the company have seen an increase in complaints from customers surrounding the robustness of the software. The company decided to implement a DevOps Pipeline (assume that this is similar to the Pipeline in this coursework but with the addition of application specific testing) but have not considered any other aspects of DevOps. This Pipeline alone has not made any difference to the robustness of the software that the customers receive and the company are confused by this. You are asked to analyse the business and identify key challenges that are preventing robust software delivery. You should then make specific recommendations as to how the company can overcome these challenges. Assume that the company currently operate with independent Developers, Operations and Quality Assurance teams. You should not discuss technological solutions to challenges here and should focus on the cultural aspects of DevOps.

Use the following citation format Harvard (author–date) style in your written component:

As stated by Shahin et al. (Shahin, 2017) or as stated in [1]

Then include your reference under references section in your report following the format below.

[1] Shahin, M., Babar, M.A. and Zhu, L., 2017. 'Continuous integration, delivery and deployment: a systematic review on approaches, tools, challenges and practices'. *IEEE Access*, 5, pp.3909-3943.

1.3 Part 3- Weekly Meeting Notes

Provide weekly meeting discussion and notes along with attendance records. Explain each member's contribution and progress each week.

Week No.	Meeting Date & time	Attendees	Weekly Progress by team member

2 Practical Component- 40 Marks

2.1 Guidance

For the practical element of this coursework, you are asked to implement a DevOps Continuous Delivery pipeline as outlined in **Figure 1** .

You should complete the Preliminary Steps **(see Preliminary Steps)** before proceeding. The table below **(Practical Tasks)** provides an outline of the steps required to complete the practical element of this coursework along with the marks available for each component. The code for this coursework is provided in Practical Code section**.**

Marks indicated are for a complete working solution to the task and proportionate marks will be awarded for partial solutions.

Upon completion of these tasks, you should record a demonstration video of this pipeline following the guidance **(see Practical Evidence, Video Demonstration)** in this specification.

Please read section Practical Evidence carefully before completing the practical components of this Coursework.

2.2 Preliminary Steps

For this coursework you will should;

a) Create two EC2 Instances on AWS with the following parameters:

Instance 1

- Name = Build Server
- Size = t2.large
- OS = Ubuntu
- Storage = 12GB
- Ports = 8080 open

Instance 2

- Name = Production Server
- Size = t2.large
- OS = Ubuntu
- Storage = 12GB

a) Prepare for the practical Coursework task by:
   - Appropriately configuring Git and GitHub authentication on your Build Server
   - Adding the **server.js file** containing the practical code **(see Practical Code)** to your local and remote git repositories
   - Installing Docker and Jenkins on the Build Server
   - Install Ansible on the production server

2.3 Practical Tasks

The table below provides an outline of the steps required to complete the practical element of this coursework along with the marks available for each component. Marks indicated are for a complete working solution to the task and proportionate marks will be awarded for partial solutions.

Task	Description	Marks
1	**Building the Application** (a) Create an appropriate Dockerfile for the code you have been provided to create a node-js web application and push this to GitHub. (b) Use this Dockerfile to manually create a Docker Image for this project (called cw2-server with version 1.0) and push this to DockerHub.	**(a) 2** **(b) 2**
2	**Preparing the Production Server** On the Production Server, write and execute Ansible playbook(s) to; (a) Install Kubectl (b) Install and Start Minikube (c) Deploy your image from DockerHub to Kubernetes (d) Create a Service for your application (e) Scale your deployment to allow for Rolling Updates to be implemented in the future. The number of Pod replicas should be chosen according to the criteria of rolling updates (see Kubernetes lecture slides) **This section should be completed with the use of Ansible.**	**(a) 4** **(b) 2** **(c) 2** **(d) 2** **(e) 2**
3	**Configuring CD Pipeline** Install and appropriately configure Jenkins to automatically: (a) Detect changes to your GitHub Repository (b) Build a Docker Image from your Dockerfile (c) Test that a Container can be launched from your Image (i.e. Build Test – e.g. run the container then run a command inside the container to ensure that it has launched successfully. You do not need to do any functional testing here) (d) Push your Image to DockerHub (e) Deploy passed builds to Kubernetes without disrupting service **This section must be completed using a Jenkins Pipeline Project and associated Jenkinsfile.**	**(a) 2** **(b) 4** **(c) 6** **(d) 6** **(e) 6**

2.4 Practical Code

server.js

```
var http = require('http');
var requests=0;
var podname= process.env.HOSTNAME;
var startTime;
var host;

var handleRequest = function(request, response) {
  response.setHeader('Content-Type', 'text/plain');
  response.writeHead(200);
  response.write("DevOps Coursework 2! | Running on: ");
  response.write(host);
  response.end(" | v=0\n");
  console.log("Running On:" ,host, "| Total Requests:", ++requests,"| App
Uptime:", (new Date() - startTime)/1000 , "seconds", "| Log Time:",new Date());}

var www = http.createServer(handleRequest);
www.listen(8081,function () {
    startTime = new Date();;
    host = process.env.HOSTNAME;
    console.log ("Started At:",startTime, "| Running On: " ,host, "\n" );
});
```

2.5 Practical Evidence

There are **FIVE** pieces of evidence required for the practical component of this Coursework which should be provided **in the evidence_YourNAME.doc** document. Failure to provide **ANY** of these may result in marks not being awarded.

1. **Docker Evidence**
   You should provide the Dockerfile that you have written and a screenshot of your DockerHub showing that the Image has been pushed.

2. **Ansible Evidence**
   You should provide a screenshot of your Terminal showing that all stages of each of your Ansible playbooks pass. If you do not provide this evidence you will not be awarded the full marks based on the playbook alone.

3. **All Ansible Playbooks**
   Include all Ansible Playbooks you have created

4. **Completed Jenkinsfile**
   Include your completed Jenkinsfile if you created one.

5. **Jenkins Console Output**
   You should include a copy of the **FULL** Jenkins Console Output for your final build. This is better copy/pasted as text than as a screenshot.

2.6 Video Demonstration

You should create a short video demonstration of your Continuous Integration Pipeline.

Your video should be no longer than 5 minutes in duration and **should be narrated** such that it is clear to the viewer that you understand the steps being carried out.

The video should be recorded **after** all of the practical tasks in **Practical Tasks** have been completed.

Before you start the video you should ensure that your pipeline is ready.

During your video you should:

- Use the 'curl $(minikube ip)' command with the node port number on the Production Server to show the initial version of the application from **Task 2** is running correctly.

- Update the code line '`response.end(" | v=0\n");`' in server.js to read '`response.end(" | v=1\n");`' and push this change to GitHub. If you have updated this value already during your development, simply increment it by 1 so that is clear that there has been a version change during your video.

- Review the full Jenkins Console Output of this build. When viewing the Console Output you should clearly indicate where there is evidence of:
    - The Docker Image being created
    - The Docker Image being pushed to DockerHub
    - The update command being issued to Kubernetes

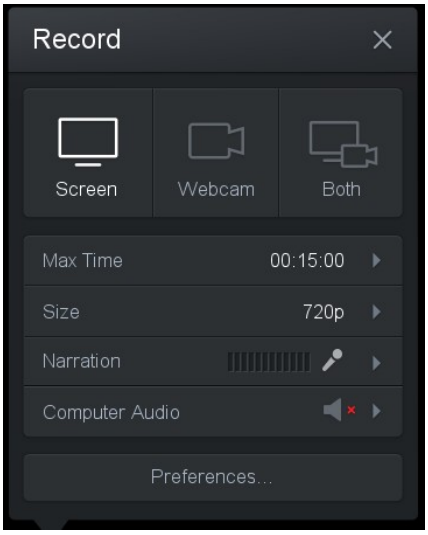
- Use the 'curl $(minikube ip)' command with the node port number on the production server to show that the new application version has been successfully deployed.
- Use the command 'kubectl get pods --watch' and show the status of the pods.

2.7 Recording Your Video

You can use the following free tool to record your video:
https://screencast-o-matic.com/screen-recorder

Click the 'Launch Free Recorder' button and follow the instructions to get started.

When the application launches you should see a control panel like this:



Select ‘Screen’ from the top three icons and ensure that your Microphone us selected under the ‘Narration’ option.

You can adjust the size of the recording area by dragging the black and white border to adjust the window size or you can set a specific size using the ‘Size’ option.

Save your video as an MP4 file (**other formats will NOT be accepted**) with the following naming convention: Group_Name_DevOps_CW1.mp4

This video should be uploaded to your GCU provided OneDrive account and a shared link (**accessible by everyone with no expiry**) to the video should be submitted to the Coursework 1 Practical Submission. Your Supplementary_Evidence.doc should be submitted to this portal along with the video - instructions will be provided with the submission portal on how to do this.

Please note that links to other cloud services WILL NOT BE ACCEPTED.

If your video is not accessible to everyone or expires it may not be available for marking/moderation and could result in loss of marks.

3 Coursework 1: Individual review report – 10 Marks

Write contributions and your teammates' contributions in details

Your contribution
I took the role of creating the … so on,

Member 2
Members' role and contributions … so on,

Member 3
Members' role and contributions … so on,

Member 4
Members' role and contributions … so on,

Member 5
Members' role and contributions … so on,

Member 6
Members' role and contributions … so on

3.1 Peer Assessment Form

	Own	Colleague 1	Colleague 2	Colleague 3	Colleague 4	Colleague 5
NAME :						
Attendance, punctuality & participation at group meetings (5)						
Production of ideas and suggestions (5)						
Performance of designated tasks (5)						
Maintaining and improving group relations (5)						
Share of group workload & overall contribution to the project (5)						

Total (25)	*0*	*0*	*0*	*0*	*0*	*0*

You must allocate marks for each group member (including yourself).

You MUST also provide the Total mark

These marks are CONFIDENTIAL – and should NOT be discussed with other students.

The marks MUST be a fair reflection on the contribution of each student

This form MUST be submitted as part of your Individual Report

Note: Consistently low marks from other group members will affect your overall CW marks.

Individual Mark

DESCRIPTION OF QUALITY		MARK RANGE
i	No contribution	0
ii	Little contribution	1 or 2
iii	Average contribution	3
iv	Considerable contribution	4
v	Outstanding contribution	5

Coursework 2 (CW2) checklist

Please provide the following checklist along with your group coursework files:

Sr. No	Component name	Confirm
1	Group based Written Component in the form of MS Word Document	☐
2	Group based Practical Component as MS Word Document including Codes and progress screenshots	☐
3	Group based Practical component video as an MP4 with the following naming convention: Group_Name_DevOps_CW2.mp4	☐
4	Individual Report as MS Word Document	☐

Appendix 1- Written Component Rubric

Coursework 1 Written Rubric	Poor (0-39%)	Basic (40-49%)	Moderate (50-59%)	Good (60-69%)	Excellent (70-100%)
Continuous Integration [20]	Insufficient or not attempted.	Some information on traditional challenges has been provided with an indication of how CI addresses these OR the CI process and implementation have been explained OR all of the above is covered but the discussion is lacking depth. No references have been provided OR references are limited in number or quality.	Clear discussion of traditional challenges and impacts. Detail also provided on process and implementation. Discussion could be clarified or expanded. Some references have been provided of reasonable quality.	Traditional challenges, impacts and how CI solves these challenges has been discussed clearly along with a clear discussion of the CI process and implementation. A good selection of appropriate references has been provided.	An excellent discussion of traditional challenges and their impact with clear links as to how CI overcomes these. The CI process is explained thoroughly with clear detail on the implementation of such a pipeline. A range of highly appropriate, quality references have been provided.
Software Tools [20]	Insufficient or not attempted.	Some detail on how the tools function has been provided OR some detail on how the tools contribute to DevOps. The discussion could be clearer or more in depth. No references have been provided OR references are limited in number or quality.	Some detail has been provided on how the tools function AND their benefits to DevOps. More detail could be provided OR the discussion could be clearer. Some references have been provided of reasonable quality.	Clear discussion of how Tools function with links to their benefits in a DevOps environment and challenges they help to address. A good selection of appropriate references has been provided.	Exceptionally clear discussion of how Tools function, clearly demonstrating how they relate to the DevOps environment with clear link to specific challenges. A range of highly appropriate, quality references have been provided.
Presentation and References [10]	Presentation and references are not to an acceptable standard.	Little attention to presentation. No use of figures. References provided but not complete/consistently formatted.	Some attention to presentation. Little/no use of figures. References are mostly complete and consistently formatted.	Well-presented and easy to read. Makes good use of figures. References are almost all consistently formatted and complete.	Excellent presentation. Makes use of highly relevant figures which complement the discussion. All references are complete and consistently formatted.